

Sheeting theorem for stable capillary minimal hypersurfaces

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GBU Geometric Analysis and PDE seminar

Joint work with Gaoming Wang (BIMSA)

Background: stable minimal hypersurface

- ▶ Let $M \subset \mathbb{R}^{n+1}$ be a smooth, properly embedded, two-sided hypersurface, M is *minimal*, if
 - $\delta|M|(\psi) = 0$ for any $\psi \in C_c^1(\mathbb{R}^{n+1}, \mathbb{R}^{n+1})$;
- $\iff$ mean curvature $H = 0$.

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$\iff$ mean curvature $H = 0$.

M is *stable minimal*, if in addition,

- $\delta^2|M|(\psi) \geq 0$ for any $\psi \in C_c^1(\mathbb{R}^{n+1}, \mathbb{R}^{n+1})$;

$\iff$ the stability inequality holds:

$$\int_M |A|^2 \varphi^2 \, d\mathcal{H}^n \leq \int_M |\nabla \varphi|^2 \, d\mathcal{H}^n,$$

for any $\varphi \in C_c^1(M)$. A is the second fundamental form of $M \subset \mathbb{R}^{n+1}$, ∇ is the gradient on M .

Compactness

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- ▶ **Answer:** Schoen-Simon-Yau 1975', C. Bellettini 2025': For $2 \leq n \leq 6$, a sequence of smooth, stable minimal *immersed* hypersurface in $\mathbb{R}^{n+1}$, with locally uniform area bounds, admits convergence subsequence, and the convergence is in the smooth topology.

Proof. Curvature estimate.

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- ▶ **Definition:** A point $X \in \overline{M}$ is called a *regular point* if there exists $\rho_X > 0$ such that

$$\overline{M} \cap B_{\rho_X}(X)$$

is a smooth, embedded hypersurface without boundary in $B_{\rho_X}(X)$. Otherwise, X is called a *singular point*. We denote the singular set by

$$\text{Sing } M \subset \overline{M}.$$

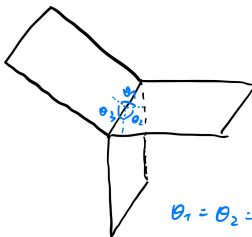
Appearance of singularities

► **Question:** what if M has singularities?

► **Answer** (in the embedded case):

Schoen–Simon 1981: assuming locally uniform area bounds and that the singular set has locally finite $(n - 2)$ -dimensional Hausdorff measure, $\implies$ **Compactness**.

The assumption on the singular set rules out classical singularities such as a **triple junction**.



$$\theta_1 = \theta_2 = \theta_3 = \frac{2}{3}\pi.$$

Sheeting theorem

Sheeting Theorem: Let $n \geq 2$, $\Lambda \in (1, \infty)$. There exists $\delta = \delta(n, \Lambda) > 0$ with the following property: For any stable minimal hypersurface $M \subset \mathbb{R}^{n+1}$ with

- ▶ $0 \in \overline{M}$;
- ▶ $\mathcal{H}^n(M \cap (B_2^n(0) \times (-2, 2))) \leq \Lambda$;
- ▶ $\mathcal{H}^{n-2}(\text{Sing} M) < \infty$,

if $E_2 \leq \delta$, then there exists $q = q(M) \in \mathbb{N}$ and smooth functions $u_i : B_1^n(0) \rightarrow \mathbb{R}$, $i = 1, \dots, q$,

$$\overline{M} \cap (B_1^n(0) \times (-1, 1)) = \bigcup_{i=1}^q \text{graph}(u_i).$$

Moreover, $u_1 < u_2 < \dots < u_q$, and

$$\|u_i - u_i(0)\|_{C^2(B_1^n(0))} \leq C(n)E_2^{\frac{1}{2}}.$$

Allard regularity theorem

Allard regularity. Let $n \geq 2$. There exists $\delta = \delta(n) \in (0, \frac{1}{2})$ with the following property.

Let $\rho > 0$, let V be a stationary integral n -varifold in $B_\rho(0)$ such that $0 \in \text{spt}\|V\|$ and

$$\frac{\|V\|(B_\rho(0))}{\omega_n \rho^n} \leq 1 + \delta.$$

Then 0 is a multiplicity-one regular point of V . In fact, there exist $\mathcal{R} \in \text{SO}(n+1)$ and $u \in C^\infty(B_\rho^n(0))$ such that

$$V \llcorner B_{\rho/2}(0) = \left| \mathcal{R}(\text{graph}(u)) \right| \llcorner B_{\rho/2}(0).$$

Allard regularity theorem

- By the monotonicity formula and upper semi-continuity of density, the assumption can be written as

$$0 \leq \frac{\|V\|(B_\rho(0))}{\omega_n \rho^n} - 1 \leq \delta,$$

the **quantity** is also known as *area excess*.

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- ▶ **Question:** what if we replace 1 by $q \in \mathbb{N}^+$?
- ▶ **Answer:** No if only assuming minimality, think of the blow-down of a catenoid.

Compactness and regularity

► **Sheeting theorem**

► **Blow-up analysis**, based on

- Federer's dimension reduction argument,
- J. Simons' uniqueness theorem for regular stable minimal cones.



Compactness and regularity

$$\dim_{\text{Haus}}(\text{Sing } M) \leq n - 7$$

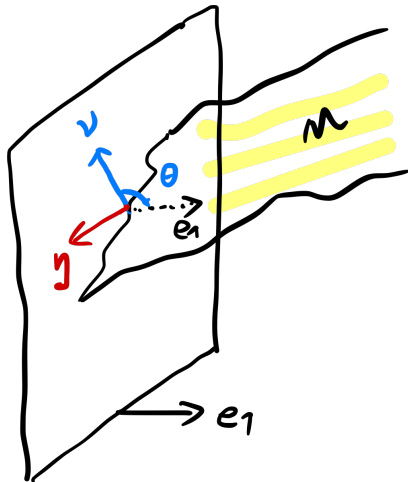
Stable minimal capillary hypersurface

- ▶ Let $\mathbb{H}^{n+1} = \{x \in \mathbb{R}^{n+1} : x_1 > 0\}$, $\theta \in (0, \pi)$.
- ▶ Let $M \subset \mathbb{H}^{n+1}$ be a smooth, properly embedded, two-sided hypersurface, with $\partial M \subset \partial \mathbb{H}^{n+1}$. M is *capillary minimal*, if
 - $H = 0$ on M and $\langle \nu, e_1 \rangle = \cos \theta$ on ∂M . M is *stable capillary minimal*, if in addition,
 - the stability inequality holds:

$$\int_M |A|^2 \varphi^2 \, d\mathcal{H}^n \leq \int_M |\nabla \varphi|^2 \, d\mathcal{H}^n + \cot \theta \int_{\partial M} A(\eta, \eta) \varphi^2 \, d\mathcal{H}^{n-1}, \quad \forall \varphi \in C_c^1(\mathbb{R}^{n+1}),$$

where η is the outer unit co-normal of $\partial M \subset M$.

Stable minimal capillary hypersurface



Capillary tilt-excess

- We introduce the *capillary tilt function*

$$g_\theta(X) := \sqrt{\underbrace{1 - \nu_1^2(X) - \nu_{n+1}^2(X)}_{\geq 0} + k(\cos \theta - \nu_1(X))^2}, \quad \forall X \in M,$$

where $k = k(n) = 1$ if $n = 2$, $= \frac{1}{n-2}$ if $n \geq 3$, and $\nu_i = \langle \nu, e_i \rangle$ with ν the unit normal vector field of M .

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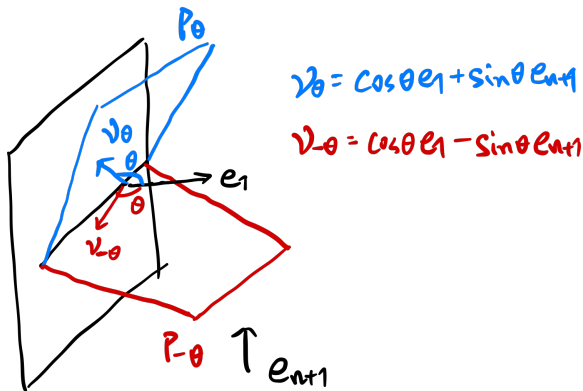
where $k = k(n) = 1$ if $n = 2$, $= \frac{1}{n-2}$ if $n \geq 3$, and $\nu_i = \langle \nu, e_i \rangle$ with ν the unit normal vector field of M .

- Clearly $g_\theta \geq 0$ and if $g_\theta(X) = 0$, then

$$\begin{cases} \langle \nu(x), e_1 \rangle = \nu_1(X) = \cos \theta, \\ 1 - \nu_1^2(X) - \underbrace{\nu_{n+1}^2(X)}_{\Rightarrow \nu_{n+1} = \pm \sin \theta} = 0, \end{cases}$$

$$\Rightarrow \nu(X) = \cos \theta e_1 + \sin \theta e_{n+1} =: \nu_\theta \quad \text{or} \quad = \cos \theta e_1 - \sin \theta e_{n+1} =: \nu_{-\theta}.$$

Toy model

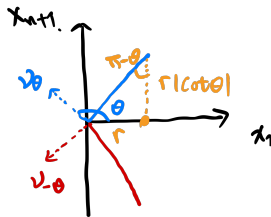


- The half-planes 1. $P_\theta \perp \nu_\theta$, $P_\theta \subset \mathbb{H}^{n+1}$; 2. $P_{-\theta} \perp \nu_{-\theta}$, $P_{-\theta} \subset \mathbb{H}^{n+1}$ are stable capillary minimal, respectively. Moreover, $g_\theta \equiv 0$ on $P_\theta, P_{-\theta}$.

Capillary tilt-excess

- define the region

$$\mathbb{C}_r^\theta := \overline{\mathbb{H}^{n+1}} \cap (B_r^n(0) \times (-(1 + |\cot \theta|)r, (1 + |\cot \theta|)r)),$$



- capillary tilt-excess (choosing $k = \frac{1}{n-2}$ in the definition of $g_{\theta,k}$):

$$E_{r,\theta} := \frac{1}{r^n} \int_{M \cap \mathbb{C}_r^\theta} g_\theta^2 d\mathcal{H}^n.$$

Sheeting theorem: capillary boundary

Sheeting Theorem: smooth case (Wang-Z' 26): Let $n \geq 2, \theta \in (0, \pi), \Lambda \in (1, \infty)$. There exists $\delta = \delta(n, \theta, \Lambda) > 0$ with the following property: For any smooth, embedded, stable capillary minimal hypersurface $M \subset \mathbb{H}^{n+1}$ with $\mathcal{H}^n(M \cap \mathbb{C}_2^\theta) \leq \Lambda$. If $E_{2,\theta} \leq \delta$, then there exists $q^\pm \in \mathbb{N}$ and smooth functions $u_i^\pm : B_1^n(0) \cap \{x_1 > 0\} \rightarrow \mathbb{R}$, $i \in Q^\pm = \{1, \dots, q^\pm\}$ ($Q^\pm$ could be empty, corresponding to the case $q^\pm = 0$),

$$M \cap \mathbb{C}_1^\theta = \left(\bigcup_{i \in Q^+} \text{graph}(u_i^+) \right) \cup \left(\bigcup_{j \in Q^-} \text{graph}(u_j^-) \right).$$

- ▶ The orienting unit normal to $\text{graph}(u_i^\pm)$ is C^0 -close to $\nu_{\pm\theta}$;
- ▶ if $q^\pm \geq 2$ then $u_i^\pm < u_{i+1}^\pm$ (i.e. ordered sheets).

L^2 -curvature estimate: Schoen inequality

Proposition (Schoen inequality): For any smooth, stable minimal hypersurface $M \subset \mathbb{R}^{n+1}$, (recalling g is the classical tilt function)

$$\int_M |A|^2 \varphi^2 \, d\mathcal{H}^n \leq C(n) \int_M g^2 |\nabla \varphi|^2 \, d\mathcal{H}^n, \quad \forall \varphi \in C_c^1(\mathbb{R}^{n+1}).$$

Proof. Testing the stability inequality with $\phi = g\varphi$, integration by parts,

$$\int_M (|A|^2 g^2 + g \Delta g) \varphi^2 \, d\mathcal{H}^n \leq \int_M g^2 |\nabla \varphi|^2 \, d\mathcal{H}^n.$$

So it suffices to bound $(|A|^2 g^2 + g \Delta g)$ from below by $C(n)|A|^2$.

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Proof. By direct computation, one obtains the Schoen differential inequality:

$$|A|^2 g^2 + g \Delta g = |A|^2 - \frac{|\nabla \nu \cdot e_{n+1}|^2}{g^2} \geq \frac{1}{n} |A|^2, \text{ on } \{g > 0\},$$

in view of an algebraic estimate using minimality:

$$\frac{|\nabla \nu \cdot e_{n+1}|^2}{g^2} \leq \left(1 - \frac{1}{n}\right) |A|^2.$$



Schoen inequality in the capillary case

Proposition (Wang-Z.' 26): For any smooth, stable capillary minimal hypersurface $M \subset \mathbb{H}^{n+1}$, $n \geq 2$, $\theta \in (0, \pi)$,

$$\int_M |A|^2 \varphi^2 \, d\mathcal{H}^n \leq C(n, \theta) \int_M g_\theta^2 |\nabla \varphi|^2 \, d\mathcal{H}^n, \quad \forall \varphi \in C_c^1(\mathbb{R}^{n+1}).$$

Schoen inequality in the capillary case

Some useful facts:

- The function g_θ satisfies,

$$\frac{\partial g_\theta^2}{\partial \eta} = -2 \cot \theta A(\eta, \eta) g_\theta^2, \quad \text{on } \partial M,$$

where η is the outer unit co-normal of $\partial M \subset M$.

Proof. By capillary boundary condition ($\implies \eta$ is a principal direction) and direct computation. □

Schoen inequality in the capillary case

Proof of Schoen inequality (capillary case): Testing the stability inequality

$$\int_M |A|^2 \phi^2 \, d\mathcal{H}^n \leq \int_M |\nabla \phi|^2 \, d\mathcal{H}^n + \cot \theta \int_{\partial M} A(\eta, \eta) \phi^2 \, d\mathcal{H}^{n-1}$$

with $\phi = g_\theta \varphi$, one gets

$$\begin{aligned} \int_M |A|^2 g_\theta^2 \varphi^2 &\leq \int_M g_\theta^2 |\nabla \varphi|^2 + \varphi^2 |\nabla g_\theta|^2 + \frac{1}{2} \langle \nabla g_\theta^2, \nabla \varphi^2 \rangle + \int_{\partial M} \cot \theta A(\eta, \eta) g_\theta^2 \varphi^2 \\ &= \int_M g_\theta^2 |\nabla \varphi|^2 + \varphi^2 \left(|\nabla g_\theta|^2 - \frac{1}{2} \Delta g_\theta^2 \right) + \frac{1}{2} \int_{\partial M} \varphi^2 \frac{\partial g_\theta^2}{\partial \eta} + \int_{\partial M} \cot \theta A(\eta, \eta) g_\theta^2 \varphi^2, \end{aligned}$$

where $\frac{\partial g_\theta^2}{\partial \eta} = -2 \cot \theta A(\eta, \eta) g_\theta^2$ on ∂M .

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with $\phi = g_\theta \varphi$, one gets

$$\int_M (|A|^2 g_\theta^2 + g_\theta \Delta g_\theta) \varphi^2 \leq \int_M g_\theta^2 |\nabla \varphi|^2.$$

So it suffices to bound $(|A|^2 g_\theta^2 + g_\theta \Delta g_\theta)$ from below by $C(n, \theta)|A|^2$.

Schoen inequality in the capillary case

► By direct computations,

$$g_\theta \Delta g_\theta = |A|^2 (k \cos \theta \nu_1 + (1 - k) \nu_1^2 + \nu_{n+1}^2) \\ - \underbrace{\left((1 - k) |\nabla \nu_1|^2 + |\nabla \nu_{n+1}|^2 + \frac{|\nabla \nu \cdot (\nu_1 e_1 + \nu_{n+1} e_{n+1} + k(\cos \theta - \nu_1) e_1)|^2}{g_\theta^2} \right)}_{:= \sum_{i=1}^3 |\nabla \nu \cdot \vec{a}_i|^2}.$$

Hence

$$(|A|^2 g_\theta^2 + g_\theta \Delta g_\theta) = |A|^2 (1 + k \cos^2 \theta - k \cos \theta \nu_1) - \sum_{i=1}^3 |\nabla \nu \cdot \vec{a}_i|^2.$$

Schoen inequality in the capillary case

Key but technical Lemma

► Define $s(n, \theta)$ by

$$s(n, \theta) = \frac{(n-1)(2 - k \sin^2 \theta) + \sqrt{4(1 - k \sin^2 \theta) + (n-1)^2 k^2 \sin^4 \theta}}{2n}.$$

Then, for all contact angles $\theta \in (0, \pi)$ and dimensions $n \geq 2$,

$$\sum_{i=1}^3 |\nabla \nu \cdot \vec{a}_i|^2 \leq s(n, \theta) |A|^2.$$

► For $\theta \in (0, \pi)$ and $n \geq 2$, we have

$$C_{n,\theta} := 1 + k \cos^2 \theta - k |\cos \theta| - s(n, \theta) > 0.$$

Schoen inequality in the capillary case

Proof continued. Hence

$$\begin{aligned}(|A|^2 g_\theta^2 + g_\theta \Delta g_\theta) &= |A|^2 (1 + k \cos^2 \theta - k \cos \theta \nu_1) - \underbrace{\sum_{i=1}^3 |\nabla \nu \cdot \vec{a}_i|^2}_{\leq s(n, \theta) |A|^2} \\ &\geq |A|^2 (1 + k \cos^2 \theta - k |\cos \theta| - s(n, \theta)) \\ &= |A|^2 C_{n, \theta}.\end{aligned}$$



Compactness and regularity in the capillary case

Remark. The capillary sheeting theorem extends to hold when **singularities** appear, with negligible $\mathcal{H}^{n-2}$ -measure.

$\implies$ Compactness & regularity for stable capillary minimal hypersurfaces.

Bernstein theorem

- Compactness + a recent result of Chodosh-Edelen-Li (uniqueness of regular stable capillary minimal cone)
⇒ Curvature estimate & Bernstein theorem (with area growth assumption).

Theorem (Wang-Z. '26) For any $\theta \in (0, \pi)$, if $M \subset \mathbb{H}^4$ is a smooth, properly embedded, complete, stable capillary minimal hypersurface, satisfying the Euclidean area growth condition, then M must be flat.

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- This result extends to hold up to $\mathbb{H}^7$, but θ is restricted.

Open questions

- ▶ Does the Bernstein theorem hold up dimension $\mathbb{H}^7$ for any $\theta \in (0, \pi)$? (In the graphical case, a very recent work by [Edelen-Li-Zhu 26'](#), $n \leq k_*(\theta)$ for $\mathbb{H}^n$)

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- ▶ By our compactness result. One could also ask, if the uniqueness result for regular stable capillary minimal cone hold up to $\mathbb{H}^7$?

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- ▶ By our compactness result. One could also ask, if the uniqueness result for regular stable capillary minimal cone hold up to $\mathbb{H}^7$?
- ▶ Stable Bernstein theorem for stable capillary minimal immersed hypersurface in the half-space in $\mathbb{H}^{n \geq 4}$?

Outreach: Stable minimal immersed hypersurfaces

- ▶ **Definition:** A point $X \in \overline{M}$ is called a (**immersed**) *regular point* if there exists $\rho_X > 0$ such that $\overline{M} \cap B_{\rho_X}(X)$ is a smooth, **immersed** hypersurface without boundary in $B_{\rho_X}(X)$. Otherwise, X is called a (**immersed**) *singular point*. We denote this singular set by $\text{Sing}_i M \subset \overline{M}$.
- ▶ If $\overline{M} \setminus \text{Sing}_i M$ is a stable minimal immersed, two-sided hypersurface in $\mathbb{R}^{n+1}$ with $\text{Sing}_i M$ of locally $\mathcal{H}^{n-2}$ -finite measure, then M is called a *branched stable minimal hypersurface*.
- ▶ In the case $n = 2$, the *generalized minimal surfaces* used to solve the Plateau problem a priori have (possibly non-empty) isolated branch points.

Multiplicity-2 class

- ▶ For a branched stable minimal hypersurface M , suppose that $\overline{M}$ has everywhere density ≤ 2 .
Wickramasekera 08' proved a regularity and compactness result for this class of hypersurfaces.
- ▶ For an immersed regular point $X \in M$ with density = 2, we have: every $C \in \text{VarTan}(V, X)$ must be a union of two distinct, mutually intersecting (multiplicity-1) planes. We call it a **cross cone**.

Sheeting theorem near cross cone

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- ▶ **Observation:** Reflecting the capillary cone C_θ in $\mathbb{H}^{n+1}$ across $\partial\mathbb{H}^{n+1}$, we obtain a cross cone, say $\hat{C}_\theta$.

Therefore, the capillary tilt function w.r.t C_θ is also the tilt function w.r.t the cross cone $\hat{C}_\theta$. And the estimates established in [Wang-Z. 26'](#) give a Schoen differential inequality w.r.t $\hat{C}_\theta$.

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Sheeting theorem near cross cone

- ▶ The Schoen differential inequality w.r.t $\widehat{C}_\theta$ is an intrinsic elliptic PDE on M .
- ▶ Using a method (De Giorgi iteration) recently developed by [Bellettini 25'](#), we can give the sheeting theorem w.r.t $\widehat{C}_\theta$ a relatively short proof.
- ▶ Since $\theta \in (0, \pi)$ is arbitrary, this covers all cross cones after rotation.

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Thank you very much for your attention!
